train_biases.py — technical

scripts/train_biases.py · 187 lines · scripts/train_biases.py

Purpose

Layer-wise bias training with cached original activations.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`main`

```
def main() -> None
```

lines 43-183 · raises RuntimeError, SystemExit · settings BIAS_BATCH_SIZE, BIAS_EVAL_INTERVAL, BIAS_LEARNING_RATE, BIAS_TRAINING_OBJECTIVE, BIAS_TRAINING_STEPS, CALIBRATION_BATCH_SIZE, CHECKPOINT_FORMAT, MODEL_DTYPE, MODEL_PATH or HF_MODEL_ID, REPLACEMENT_METHOD, SEED

Script entry point: runs this stage end to end.

1. Exit early unless `REPLACEMENT_METHOD=bias`; require `replaced-model/`; prepare calibration with the task.
2. The replaced model's manifest must be a bias replacement of the same model with the same calibration fingerprint.
3. Reference cache: if `CACHE_DIR/.../layer_reference/` has valid entries (same model, dtype, fingerprint) reuse it; otherwise load the original model, `capture_layer_references` for the pruned layers, save, free the model.
4. Load the replaced model with the configured placement; `freeze_all_but_biases`; verify the trainable set is exactly the BiasExpert biases and the routers are frozen; snapshot frozen parameters of the trained layers.
5. Seed; for each pruned layer: load its cache, map calibration ids to cached rows, `train_layer_biases`.
6. Verify every snapshotted frozen parameter is bit-identical.
7. Write `training_report.json`; disable gradients; save `trained-model/` (manifest stage `trained` + training summary) in `CHECKPOINT_FORMAT`; log per-layer loss changes.

Errors raised

Function	Exception	Message (f-string fields in braces)
`main`	SystemExit	REPLACEMENT_METHOD={cfg.replacement_method} has no trainable parameters; nothing to train. Evaluate with `scripts/evaluate.py --stage before_training`.
`main`	SystemExit	The replaced model was built with different calibration data/settings. Re-run collect_stats.py, score_experts.py and replace_experts.py.
`main`	SystemExit	The replaced model was not built with REPLACEMENT_METHOD=bias. Re-run replace_experts.py.
`main`	SystemExit	The replaced model was built from a different base model.
`main`	RuntimeError	Trainable parameters are not exactly the BiasExpert biases
`main`	RuntimeError	Frozen parameter {name} changed during training